



Technical Specifications:

- Payload capacity: 2kg
- Max speed: 120km/hr
- Ceiling: 3048m
- Max take-off weight: 21kg
- Width: 2.6m
- Length: 1.6m
- Endurance: 10hr
- Range: 643km

Fig. 1: Super Bat DA-50

disturbed when they are incubating an egg or hatching it for protection. Hence the drones used should not be noisy or emitting laser lights to avoid disturbance.

Recent research shows that traditional approaches for wildlife monitoring can be dangerous and tedious, while drones can efficiently collect precise observational data for wildlife monitoring. Based on this advantage, research is going on for wildlife monitoring with drones, including thermal monitoring during night or dense forest monitoring. Some of the best practices derived by scientists to manage situations and mitigate, or even alleviate potential physical disturbances to wildlife, when using drones include:

- Expert advice and local resources be used to study the location before using drones for monitoring.
- Appropriate approval and permissions be taken from authorities for using the drones.
- Proper training and dry run experiments be conducted in simulated environment to avoid unforeseen failures in real-time monitoring.
- Proper understanding of civil aviation rules—including restrictions in flying distance/height, border conditions, altitude guidelines, night-time monitoring guidelines—

be ensured first.

- Drones be used only for the research and not to disturb animals and threaten their regular life.
- Launch and operation of drones be managed from recoverable distance to avoid failure or accident.
- Drones should be designed to monitor behavioural and physiological stress responses, when possible, and record for study purpose.
- Be ready to abort drone operation when there is excessive disturbance caused to animals.

Drones and AI (artificial intelligence) technology are advancing and evolving wildlife surveys. Tracking species through collaring can be dangerous and may be invasive and disruptive to the animals. Gathering data via drones is more efficient and safer for both animals and the researchers. Using drones, biologists can collect the information they need through specialised sensing technology, including lidar, RGB, corona, hyperspectral, and thermal sensors. Importance of drones is recognised for environmental monitoring to track the health and population growth of indicator species and monitoring the overall health of habitats and ecosystems.

Drones for wildlife monitoring

Wildlife monitoring is a challenging activity among environmental activities like wildlife's social environment, birds, and natural resources such as volcano, waterfalls, and rivers. Since drones are equipped with cameras, laser lights, GPS tracking, altimeters, etc, they are highly effective for wildlife monitoring.

Modern conservation drones also use thermal and long-distance focal-point cameras to monitor wildlife



Technical Specifications:

- Endurance: Up to 5 hours
- Maximum service altitude: 1,220 metres
- Maximum control range: 50km
- 30km live video stream
- Daytime video: 10x zoom Sony block camera
- Night time operations: Flir 640 Vue Pro, Tamarisk 320 or 640 Thermal Imager
- Gimbal: Stabilised 2- and 3-axis with brushless motors
- On-board GPS
- Waypoint navigation
- Flight stability systems

Fig. 2: Air Shepherd ZT-TIC